Rakuten Image Classification - Colab Fine-Tuning

This standalone notebook fine-tunes an ImageNet-pretrained ResNet18 on the Rakuten product images. It keeps the complete shared validation set unchanged, reports weighted and macro F1 after every epoch, and saves the best checkpoint by validation macro F1.

The confidential images and manifests must remain in a private Google Drive folder and must not be committed to GitHub.

1. Configuration

The schedule first warms up the new classification head and then fine-tunes ResNet's final residual block (`layer4`). Full-network fine-tuning is evaluated separately in `03_3_image_resnet18_full_finetuning.ipynb` .

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

2. Stage images and verify the shared split

Training rows may have been cleaned for exact duplicate leakage. Validation must contain all 16984 products from the shared split so that text and image scores remain directly comparable and suitable for fusion.

	rows	classes
train	62312	27
validation	16984	27

3. Image datasets and transformations

Training receives light stochastic augmentation. Validation uses the deterministic preprocessing supplied with the pretrained ImageNet weights.

```
Training batches  : 974
Validation batches: 266
```

4. Model, phased unfreezing and metrics

The head is warmed up first. Next, `layer4` and the head are trained with separate learning rates. Frozen BatchNorm layers remain in evaluation mode.

5. Fine-tune and retain the best validation checkpoint

Validation is evaluated after every epoch. The checkpoint selection criterion is macro F1 because it gives every class equal importance. Training loss may be higher than validation loss because only training uses stochastic augmentation.

```
Phase: head | trainable parameters: 13851
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	head	1	1.8624	1.6680	0.5042	0.4863	0.4317

```
New best checkpoint: macro F1 = 0.4317
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	2	head	2	1.6548	1.6399	0.5115	0.4956	0.4472

```
New best checkpoint: macro F1 = 0.4472

Phase: layer4 | trainable parameters: 8407579
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	3	layer4	1	1.5023	1.3989	0.5856	0.5766	0.5232

```
New best checkpoint: macro F1 = 0.5232
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	4	layer4	2	1.2000	1.3124	0.6135	0.6084	0.5723

```
New best checkpoint: macro F1 = 0.5723
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	5	layer4	3	1.0137	1.3412	0.6151	0.6115	0.5792

```
New best checkpoint: macro F1 = 0.5792
```

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	6	layer4	4	0.8506	1.3729	0.6250	0.6238	0.5902

```
New best checkpoint: macro F1 = 0.5902
```

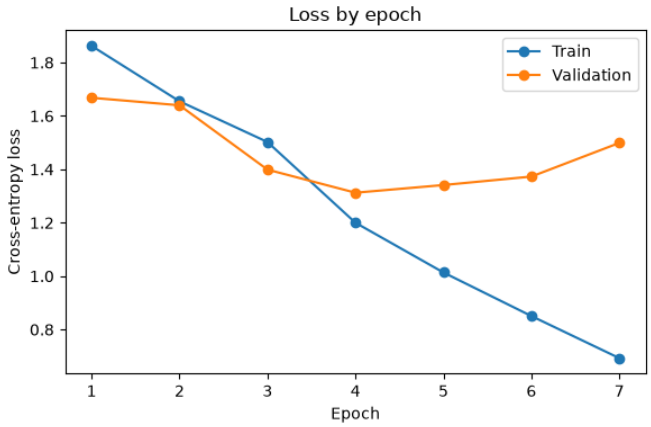
	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	7	layer4	5	0.6927	1.4990	0.6113	0.6064	0.5714

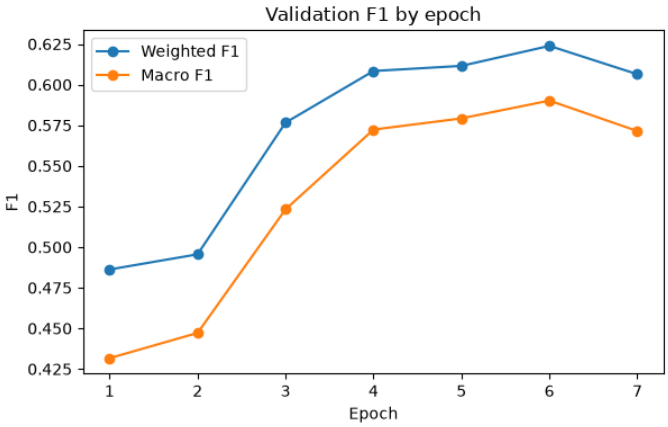
```
Best validation macro F1: 0.5902
```

6. Training curves

Use validation loss and macro F1 to assess generalisation. A falling training loss combined with a rising validation loss indicates overfitting.

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	head	1	1.8624	1.6680	0.5042	0.4863	0.4317
1	2	head	2	1.6548	1.6399	0.5115	0.4956	0.4472
2	3	layer4	1	1.5023	1.3989	0.5856	0.5766	0.5232
3	4	layer4	2	1.2000	1.3124	0.6135	0.6084	0.5723
4	5	layer4	3	1.0137	1.3412	0.6151	0.6115	0.5792
5	6	layer4	4	0.8506	1.3729	0.6250	0.6238	0.5902
6	7	layer4	5	0.6927	1.4990	0.6113	0.6064	0.5714





7. Final evaluation and fusion-ready outputs

The best checkpoint is evaluated on the complete shared validation set. Predictions retain `row_id` and all 27 class probabilities so they can later be joined with text-model outputs.

	validation_loss	accuracy	weighted_f1	macro_f1
resnet18_finetuned	1.3729	0.6250	0.6238	0.5902

	prdtypecode	precision	recall	f1	support
8	1281	0.401	0.244	0.303	414
24	2585	0.432	0.347	0.385	499
7	1280	0.368	0.448	0.404	974
6	1180	0.545	0.359	0.433	153
11	1302	0.489	0.410	0.446	498
2	50	0.388	0.530	0.448	336
22	2582	0.489	0.419	0.451	518
17	2220	0.500	0.448	0.473	165
12	1320	0.506	0.492	0.499	648
16	2060	0.514	0.492	0.503	999
0	10	0.524	0.602	0.561	623
13	1560	0.507	0.628	0.561	1015
20	2462	0.686	0.493	0.574	284
1	40	0.643	0.534	0.583	502
10	1301	0.679	0.578	0.624	161
15	1940	0.798	0.540	0.644	161
3	60	0.680	0.614	0.646	166
19	2403	0.636	0.657	0.646	955
4	1140	0.641	0.657	0.649	534
21	2522	0.696	0.649	0.672	998
9	1300	0.650	0.726	0.686	1009
18	2280	0.676	0.788	0.727	952
25	2705	0.839	0.679	0.751	552
26	2905	0.690	0.833	0.755	174
14	1920	0.784	0.779	0.782	861
23	2583	0.815	0.810	0.813	2042
5	1160	0.937	0.896	0.916	791

Interpretation checklist

Compare the fine-tuned `layer4` model with the frozen ResNet18 on exactly the same 16984 validation products. Report validation loss, accuracy, weighted F1, macro F1 and F1 per class. Full-network fine-tuning starting from this `resnet18_finetuned_best.pt` checkpoint is evaluated separately in `03_3_image_resnet18_full_finetuning.ipynb`.